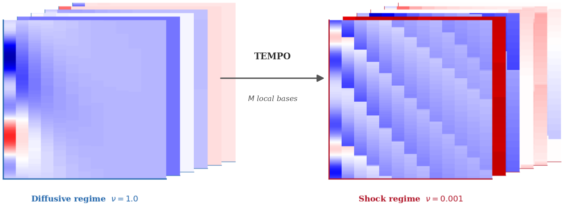
TEMPO: Regime-Aware Operator Learning with Locally Adapted Bases



Regime 1, $\nu = 1.0$: smooth, diffusive
Regime 2, $\nu = 0.001$: sharp shock front

Operator learning problem:

$$\mathcal{G} : \mathcal{U} \times \mathcal{K} \to L^2(\Omega \times \mathcal{T}), \quad (u, \kappa) \mapsto s$$

Challenge

A single global POD basis cannot efficiently represent solutions that change character fundamentally across κ.

TEMPO: discover M regimes via EM-GMM, build a dedicated POD basis per regime, route at inference:

$$\hat{s}(x; u, \kappa) = \sum_{m=1}^{M} \underbrace{w_m(u, \kappa)}_{\text{GatingNet}} \left[\Phi_0^{(m)} + \sum_{k=1}^{P_m} \underbrace{b_{m,k}(u, \kappa)}_{\text{BranchNet}_m} \Phi_k^{(m)}(x; \kappa) \right]$$